

A Meta-analysis of Central Corneal Thickness Changes With Overnight Orthokeratology

Fang Li, M.D., Zhi-Xin Jiang, M.D., Peng Hao, M.D., and Xuan Li, M.D., Ph.D.

Objectives: To assess the change of central corneal thickness (CCT) in the treatment of orthokeratology in patients with myopia.

Methods: A systematic search of all relevant studies published through April 2014 was conducted, and 95% confidence intervals (CI) of CCT change were calculated. Random or fixed-effects models were used according to heterogeneity. Publication bias of the articles was evaluated using funnel plots and Begg test.

Results: A total of 10 studies with 239 patients (339 eyes) from clinical studies were included. Central corneal thickness reduced significantly from 1 day to 1 week by 5.73 μm (95% CI, 1.75–9.70 μm ; $P=0.005$), and a significant mean reduction of 5.89 μm also occurred from 1 day to 1 month (95% CI, 3.50–8.29 μm ; $P<0.001$). No significant reduction was found between 1 week and 1 month ($P=0.32$).

Conclusions: Our meta-analysis demonstrated that most reduction of CCT occurred during the first week and remained thinner for 1 month. Further randomized controlled trials with larger sample sizes, standardized outcome measurements, and different follow-up periods are warranted to find the precise change.

Key Words: Central corneal thickness—Orthokeratology—Meta-analysis.

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Orthokeratology (OK) is a clinical technique that uses specially designed rigid gas-permeable contact lenses to reshape the cornea to temporarily modify, reduce, or eliminate refractive errors and astigmatism to achieve a transient improvement in unaided visual acuity.^{1,2} Nowadays, OK is usually regarded as an effective treatment of myopia, not as visual aids.³ As it is becoming more and more popular, concerns about safety and adverse effects have become a very hot issue. Efficacy and safety of this option not only relate to correct lens fitting and good patient compliance but also depend on regular monitoring of corneal changes.^{4,5} Central corneal thickness (CCT), which is also very important for the planning of refractive surgery and diagnosis of glaucoma, is being widely

studied to monitor the retardation of myopia and OK-related complications, such as regression, keratectasia, and so on.^{6–8}

Different studies have demonstrated different CCT changes through the programmed application of rigid contact lenses. El-Hage¹ reported in a study that no change was found in CCT during the overnight wearing of OK, whereas Mitsui and Yamada⁹ found an increase in CCT. Therefore, it is not entirely clear when and how OK affects CCT and whether short-term wear and long-term wear produce the same change.

Our meta-analysis was performed using existing comparative studies of OK wearers in an attempt to detect any difference in CCT change. To the best of our knowledge, this is the first meta-analysis concerning CCT change between different follow-up visits.

MATERIALS AND METHODS

Search Strategy

We performed a systematic search of Ovid, EMBASE, MEDLINE, and ScienceDirect to identify published reports of clinical trials comparing CCT change after overnight OK for patients with myopia. Searches were restricted to English language reports published before April 30, 2014. The keywords were “corneal thickness” or “corneal pachymeter,” “orthokeratology,” “ortho-k,” or “OK lens.” In addition, the reference lists of original reports and review articles retrieved through the search were reviewed for additional studies.

Inclusion and Exclusion Criteria

Articles from peer-reviewed medical journals were included if they reported on studies meeting the following criteria: the subjects should be patients who had a low to moderate amount of myopia (-0.75 to -6.0 diopter [D]) and with or without with-the-rule corneal astigmatism of up to 1.5 D; the follow-up period should include at least 1 day; sufficient data should be provided to gain or calculate mean value and standard deviation of CCT. Central corneal thickness in our study refers to total cornea thickness averaged centrally over a circular area of 2 mm or obtained across a horizontal meridian from specific corneal center locations by equation or visually. Studies relating to high myopia or hyperopic OK were excluded. We also excluded studies focused only on the central epithelial or stromal thickness.

Study Selection

Two reviewers (F.L. and Z.-X.J.) independently screened the titles and abstracts for the eligibility criteria. Subsequently, the full text of the studies that potentially met the inclusion criteria were read, and the literature was reviewed to determine the final

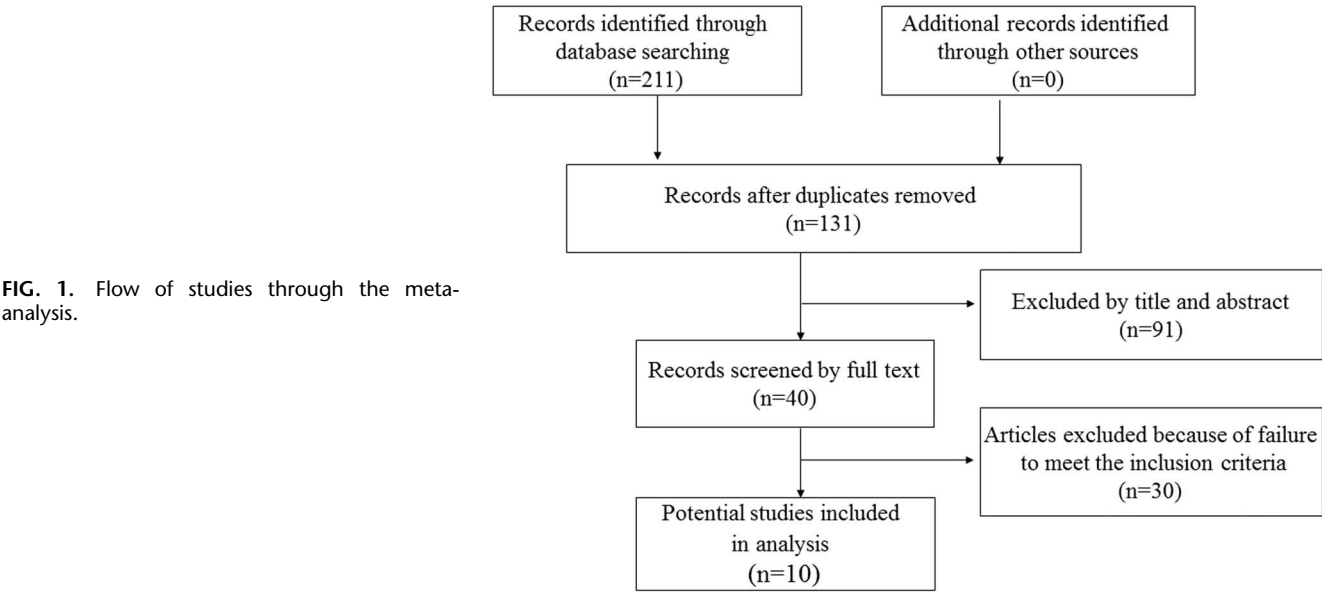
From the Tianjin Key Laboratory of Ophthalmology and Vision Science (F.L., Z.-X.J., P.H., X.L.), Tianjin Eye Hospital, Tianjin Eye Institute, Tianjin, China; and Medical School of Nankai University (F.L.), Tianjin, China.

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Address correspondence to Xuan Li, M.D., Ph.D., Tianjin Eye Hospital, No. 4 Gansu Road, Heping District, Tianjin 300020, China; e-mail: xuanli08@yahoo.com

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inclusion. Results were compared, and any discrepancy between the reviewers was resolved by consensus.

Data Extraction and Risk for Bias Assessment

For each study, the following data were extracted: titles, authors, year of publication, location of the trial, number of subjects/eyes included in the statistic analysis, patient age, sex, and the baseline spherical equivalent (in diopters). We assessed sources of systematic bias and the quality of the trials according to the methods described by Jadad et al.¹⁰ A funnel plot and Begg bias test were performed both visually and statistically to measure the publication bias with statistical significance set at $P<0.05$.

Data Synthesis and Analysis

As CCT is a type of continuous variance, we calculated the weighted mean difference. I^2 was used to quantify the inconsistency among the results of studies. A value of 25% corresponds to low, 50% to moderate, and 75% to high heterogeneity.¹¹ The Mantel-Haenszel method for fixed effects and the Der Simonian and Laird method for random effects were used to estimate pooled effects.¹² The fixed-effect model was used to pool the data when the heterogeneity was not significant. Data were shown as mean difference and 95% confidence interval (CI). Statistical significance was set at $P<0.05$ (2-tailed). Sensitivity analysis was performed to

verify the effect of single study on the stability of the summary estimates by excluding every study. The statistical analysis was performed with the use of RevMan software (version 5.0; Cochrane Collaboration, Oxford, United Kingdom).

RESULTS

Search Results

A total of 40 full texts were reviewed, of which 10 published studies satisfied the eligibility criteria.^{13–22} Flow chart of studies during meta-analysis was shown in Figure 1. All of the included studies investigated the CCT change after overnight OK for at least 1 day.

Study Characteristics

In total, 10 studies with 239 total patients (339 eyes) were eligible for inclusion. Of the enrolled 10 studies, 4 studies compared the change of CCT between day 1 and 1 week,^{13,20–22} and 4 studies compared CCT change between day 1 and 1 month.^{13,14,19,21} Characteristics of included studies were presented in Table 1. To unify the data form to help data processing, change in CCT from the baseline in studies by Nichols et al.,¹³ Chang et al.,¹⁵ and Stillitano et al.²⁰ was obtained by statistic formula because they just provided specific CCT value. Therefore, we added analysis without these three studies to exclude the impact of calculation.

TABLE 1. Characteristics of the Included Studies Evaluating the CCT in Patients With Low to Moderate Myopia

Study	Year	Country	Patients/Eyes	Mean Age (yr)±SD (Range)	Gender M/F	Spherical Equivalent (D)
Nichols et al. ¹³	2000	United States	8/8	25.9±3.9	4/4	NG
Alharbi and Swarbrick ¹⁴	2003	Australia	18/18	22–29	12/6	−2.63±0.68
Chang et al. ¹⁵	2013	China	34/34	29.65±4.35 (21–37)	14/20	−2.88±0.93
Cheung and Cho ¹⁶	2013	China	37/37	9 (7–10)	19/18	−2.29±0.74
Ishida et al. ¹⁷	2011	Japan	45/90	27.2±5 (20–37)	16/29	−2.87±1.05
Yeh et al. ¹⁸	2013	United States	39/78	22.5±4.8 (18–36)	18/21	NG
Owens et al. ¹⁹	2004	New Zealand	20/19	17–37	4/16	−2.28±0.84
Stillitano et al. ²⁰	2007	Brazil	14/26	31±8.43	7/7	−2.24±0.98
Swarbrick et al. ²¹	1998	Australia	6/11	21–27	0/6	−2.18±0.94
Yoon and Swarbrick ²²	2013	Australia	18/18	25.1±3.7 (19–32)	7/11	−2.64±0.99

CCT, central corneal thickness; D, diopter; F, female; M, male; NG, not given; SD, standard deviation.

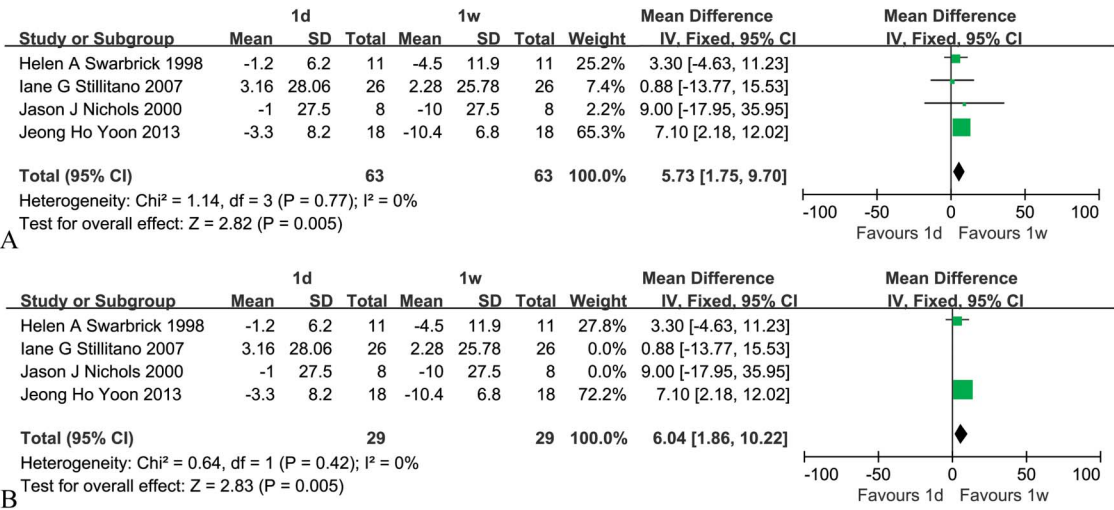


FIG. 2. Forest plot shows the change of central corneal thickness (CCT) from baseline between day 1 and 1 week. Mean value and 95% confidence interval (CI) of CCT change were shown in all the included studies (A) and studies without Nichols et al. and Stillitano et al. (B). For the calculation of SD.

Quantitative Data Synthesis

Change in CCT Between Day 1 and 1 Week

The results (Fig. 2A) showed that the CCT reduced significantly ($P=0.005$) by $5.73\text{ }\mu\text{m}$ from day 1 to 1 week, with 95% CI: 1.75 to 9.7. Sensitivity analysis was performed by excluding every study. When studies by Stillitano et al.²⁰ and Nichols et al.¹³ were excluded, P value showed no change ($P=0.005$) (Fig. 2B). When a study by Yoon and Swarbrick²² was excluded, P value turned to 0.36, which demonstrated that this study contributed a great deal to the overall result. The reasons for this influence were carefully analyzed. First, we found that this study is of high quality and scored high on the Jadad scale. In addition, calculation of standard deviation (SD) in the above two studies promoted the high weight

of the study by Yoon and Swarbrick,²² resulting in its decisive role in the overall effect.

Change in CCT Between Day 1 and 1 Month

The results (Fig. 3A) demonstrated that a significant CCT reduction of $5.89\text{ }\mu\text{m}$ occurred from day 1 to 1 month, with 95% CI: 3.5 to 8.29 ($P<0.001$). When a study by Nichols et al.¹³ was excluded, the reduction was still significant (Fig. 3B). Sensitivity analysis also showed no change in the overall results (all $P<0.05$).

Change in CCT Between 1 Week and 1 Month

The results (Fig. 4A) showed that there was no significant CCT reduction from 1 week to 1 month, with 95% CI: -2.79 to 8.5

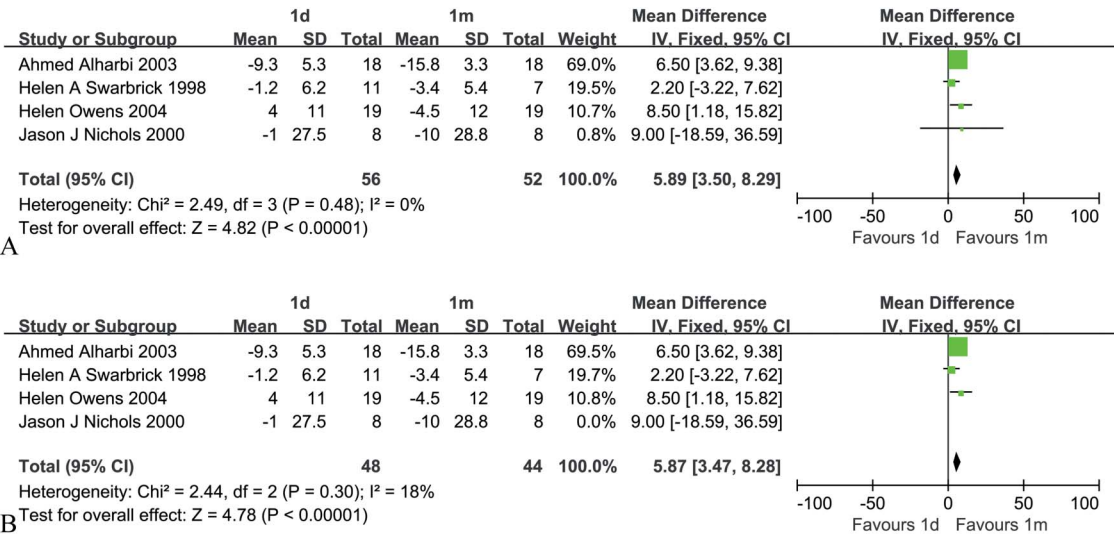


FIG. 3. Forest plot shows the change of central corneal thickness (CCT) from baseline between day 1 and 1 month. Mean value and 95% confidence interval (CI) of CCT change were shown in all the included studies (A) and studies without Nichols et al. (B) because of calculation of standard deviation (SD).

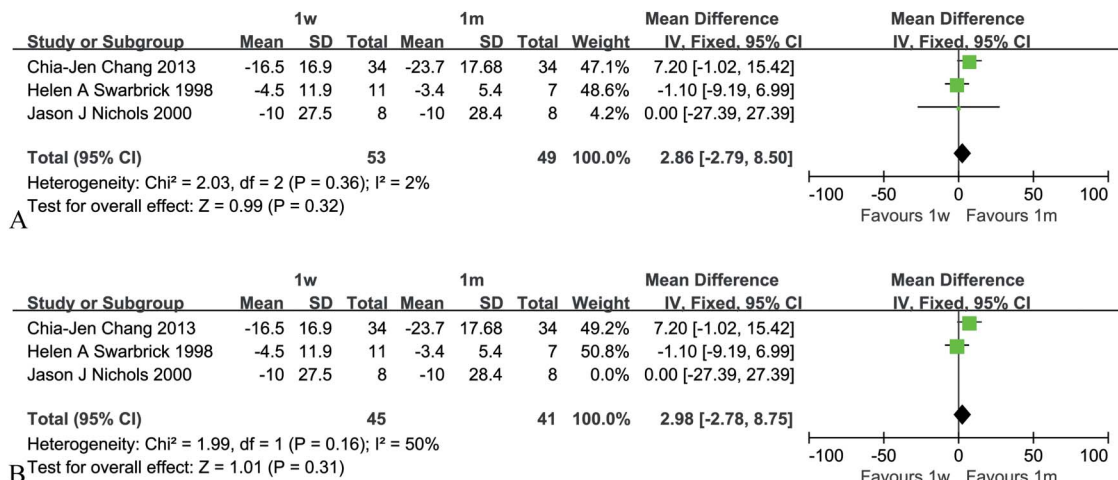


FIG. 4. Forest plot shows the change of central corneal thickness (CCT) from baseline between 1 week and 1 month. Mean value and 95% confidence interval (CI) of CCT change were shown in all the included studies (A) and studies without Nichols et al. (B) for the calculation of standard deviation (SD).

($P=0.32$). When a study by Nichols et al.¹³ was excluded, no significant reduction was showed ($P=0.47$) (Fig. 4B). Sensitivity analysis was performed, and the overall result was similar without statistical difference (all $P>0.1$).

Change in CCT Between 1 and 2 Weeks

The results (Fig. 5A) showed no significant CCT reduction from 1 to 2 weeks, with 95% CI: -0.83 to 6.2 ($P<0.001$). Exclusion of the studies by Chang et al.¹⁵ and Nichols et al.¹³ for calculations of SD showed no significant difference in P value ($P=0.29$) (Fig. 5B). Every study was excluded to perform the sensitivity analysis, and overall result were similar with significant difference (all $P<0.05$).

Change in CCT Between Baseline and 6 Months

Only 2 studies reported CCT change after 6 months of OK wear.^{16,17} Cheung et al.¹⁶ found that CCT reduced significantly by $-9\pm9\text{ }\mu\text{m}$ using Pentacam, an anterior segment tomography. Ishida et al.¹⁷ also found significant CCT reduction from baseline ($536.2\pm39.6\text{ }\mu\text{m}$) to ($528.3\pm40.0\text{ }\mu\text{m}$) in 6 months with SP-3000 pachymeter. Because of the small data set, a pooled result was unrepresentative.

Publication Bias

The publication bias test was performed separately according to different time points. No significant publication bias was shown for CCT change between 1 day and 1 week, 1 day and 1 month,

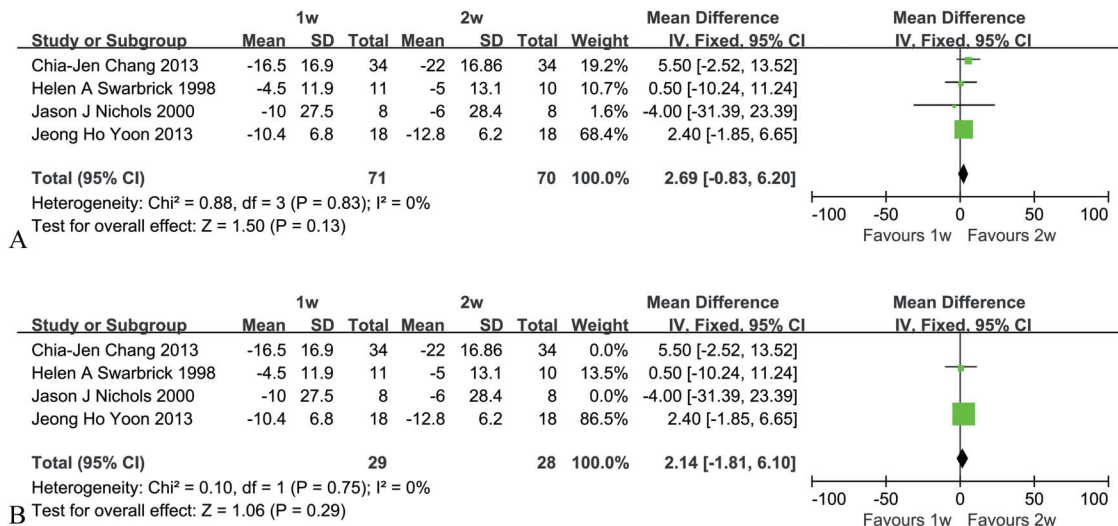


FIG. 5. Forest plot shows the change of central corneal thickness (CCT) from baseline between 1 and 2 weeks. Mean value and 95% confidence interval (CI) of CCT change were shown in all the included studies (A) and studies without Nichols et al. and Chang et al. (B) because of the calculation of standard deviation (SD).

1 week and 1 month, or 1 week and 2 weeks by Begg rank correlation method, all *P* values were larger than 0.05 (*P*=1.00, 0.308, 1.00, 0.089, respectively).

DISCUSSION

Our results demonstrated that the CCT reduction with OK mostly occurred within 1 week and remained stable for 1 month.

The cornea consists of three main layers, including the epithelium, stroma, and endothelium. As for the reason for the reduction in CCT, we should analyze the thickness changes in the epithelium and stroma because endothelium is so thin a cellular monolayer that the thickness change can be ignored.

On one hand, it is well documented in the literature that epithelial thinning occurs after OK wearing. The epithelium is the most malleable structure with a low elastic modulus, which means it is easy to deform.²³ Using optical pachometry, Alharbi and Swarbrick¹⁴ found central epithelial thinning ($-8.7 \pm 4.8 \mu\text{m}$) accounted for almost all the CCT thinning ($-9.3 \pm 5.3 \mu\text{m}$). Wang et al.²⁴ demonstrated a significant epithelial thinning in the central cornea after the use of overnight OK detected by optical coherence tomography. Reinstein et al.²⁵ also demonstrated central epithelial thinning using very high-frequency digital ultrasound arc-scanners.

With respect to the mechanism of epithelial thinning, Greenberg and Hill²⁶ found a reduced number of cell layers through histological studies of rabbits in 1973. Then, Cheah et al.²⁷ found epithelial thinning to be caused by focal cell compression rather than cell migration or the loss of cell layers through histological analysis of primates. Choo et al.²⁸ also held the view that epithelial cell compression and deformation was responsible for the thinning of epithelium, using cats as an animal model. Further investigation is needed to find out the exact mechanism in the human cornea.

On the other hand, central stromal thickness change is not explicitly known because different studies have given different results. Alharbi et al.²⁹ reported an inhibition in central corneal stromal edema response with OK, as the uneven postlens pressure may exert a “clamping” effect on the central cornea and thus lead to thinner central stromal components. In this way, the reduction in CCT was epithelial and stromal in origin. But, several studies also reported central stromal thickening. Zhong et al.³⁰ found central stromal thickness increased significantly after 8 hr of OK lens wear. Reinstein et al.²⁵ also found central stromal thickening in patients with mild myopia, but they demonstrated that the magnitude of the stromal thickening was less than one-third of the magnitude of the epithelial reduction. In this case, epithelial thinning is also responsible for the total reduction of CCT.

Our findings that CCT reduced significantly during the first week confirmed the results of the studies by Mao et al.³¹ and Mika et al.³² However, Fan et al.³³ reported most of the reduction occurred during the first 2 weeks of lens wear, whereas Alharbi and Swarbrick¹⁴ found most CCT reduction occurred after the first night and rapidly progressed to stabilization by day 10, and Soni et al.³⁴ found that 70% of the change occurred during the first day. Jayakumar and Swarbrick³⁵ stated that thinning of the central cornea occurred within 1 hr of eye opening after OK lens wear in all age groups. In addition, Lu et al.²³ reported that the change occurred 15 min after the OK.

In our meta-analysis, we did not get a pooled result regarding the difference between baseline and day 1 because of a lack of data.

Only two of six studies provided the baseline value of CCT and day 1 thickness after OK.^{13,21} Four other studies were excluded because baseline values were not provided or could not be obtained. Further work is needed to determine exactly when this change in CCT occurs.

As for the CCT change over 6 months, 2 included studies both showed significant reduction, but other studies showed different results.^{1,36} El-Hage et al.¹ did not detect any significant change in CCT over time. Nieto-Bona et al.³⁶ also reported no significant difference in CCT of 3, 6, and 12 months with baseline using confocal microscopy. As few studies document CCT change in 6 months and even fewer provide needed data, further investigation and assessments are required to evaluate long-term effects.

Yet, the thinning of CCT led to questions of whether the reduction could cause ocular surface infection. The prevalence and incidence of complications associated with overnight OK have not been determined. Until now, only some case reports and case series regarding *Pseudomonas aeruginosa* infection, *Acanthamoeba* infection, or corneal ulcers have been reported.^{37–39} A recent report by Bullimore et al.⁴⁰ showed that the incidence of microbial keratitis associated with overnight OK is similar to that of other overnight modalities. To some extent, it can eliminate the concern of adverse effects when choosing OK over other overnight modalities.

However, several limitations existed in our study. First, different measurements were used to evaluate CCT, and the effects of instrument evolution in precision or patient compliance during the evaluation are not negligible. Second, the statistical efficacy may be inadequate because the amount of studies included is finite. The limited number of studies or the limited sample size of the pooled results may influence the stability of the results. In addition, different studies chose different time points to observe changes in CCT, whereas our meta-analysis was limited to the time points of 1 day, 1 week, and 1 month. The change in CCT within 1 day is still unclear.

In conclusion, the present meta-analysis demonstrated that most of the reduction in CCT occurred during the first week and remained thinner for 1 month. Overnight OK is a relatively safe procedure for patients with myopia. Further studies with greater sample size and multiple time points are required to determine the significance of the CCT change during the wearing of overnight OK.

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